

THM – Crack the Hash

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Objective

This writeup documents the process used to solve the *Crack the Hash* room on TryHackMe. The workflow combined online hash identification and lookup with local cracking tools, primarily `john` and `hashcat`. Screenshots are included at the point where each step was performed.

1 Approach

For both tasks, the same overall methodology was used:

1. Identify or test the hashes with CrackStation to quickly recover any common unsalted values.
2. Determine the hash type for anything not immediately recognized or cracked.
3. Save the remaining hashes into local text files.
4. Use the appropriate cracking tool and mode based on the hash format.

2 Task 1

2.1 Hashes Provided

```
48bb6e862e54f2a795ffc4e541caed4d
CBFDAC6008F9CAB4083784CBD1874F76618D2A97
1C8BFE8F801D79745C4631D09FFF36C82AA37FC4CCE4FC946683D7B336B63032
$2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNkJkUlypTsgbX1H68wsRom
279412f945939ba78ce0758d3fd83daa
```

2.2 Step 1: Check Known Hashes in CrackStation

The first step was to submit the hashes to CrackStation to see whether any of them were already present in its database. This immediately recovered four of the five hashes.

Enter up to 20 non-salted hashes, one per line:

```
48bb6e862e54f2a795ffc4e541caed4d
CBFDAC6008F9CAB4083784C8D1874F76618D2A97
1C8BFE8F801D79745C4631D09FFF36C82AA37FCACCE4FC946683D7B336B63032
$2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNk3kUlypTsgbX1H68wsRom
279412f945939ba78ce0758d3fd83daa
```

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
48bb6e862e54f2a795ffc4e541caed4d	md5	48bb6e862e54f2a795ffc4e541caed4d
CBFDAC6008F9CAB4083784C8D1874F76618D2A97	sha1	CBFDAC6008F9CAB4083784C8D1874F76618D2A97
1C8BFE8F801D79745C4631D09FFF36C82AA37FCACCE4FC946683D7B336B63032	sha256	1C8BFE8F801D79745C4631D09FFF36C82AA37FCACCE4FC946683D7B336B63032
\$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNk3kUlypTsgbX1H68wsRom	Unknown	Unrecognized hash format.
279412f945939ba78ce0758d3fd83daa	md4	279412f945939ba78ce0758d3fd83daa

Color Codes: **Green** Exact match, **Yellow** Partial match, **Red** Not found.

Figure 1: CrackStation results for Task 1

From this result:

- The first, second, third, and fifth hashes were recovered.
- The fourth hash was not found and required further analysis.

2.3 Step 2: Identify Remaining Hash

\$2y\$12\$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNk3kUlypTsgbX1H68wsRom

This hash was identified as **bcrypt (Blowfish)**.

2.4 Step 3: Crack Using John

Save the hash:

```
echo '$2y$12$Dwt1BZj6pcyc3Dy1FWZ5ieeUznr71EeNk3kUlypTsgbX1H68wsRom' > hash.txt
```

Run John:

```
john --format=bcrypt --mask='?1?1?1?1' hash.txt
```

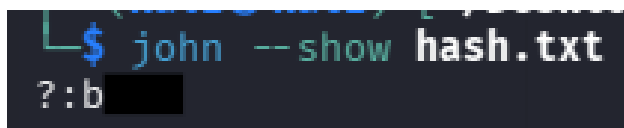


Figure 2: John cracking bcrypt hash

2.5 Task 1 Summary

- CrackStation recovered most hashes instantly.
- John the Ripper was used for the remaining bcrypt hash.

3 Task 2

3.1 Hashes Provided

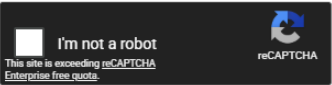
```
F09EDCB1FCEFC6DFB23DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85
1DFECA0C002AE40B8619ECF94819CC1B
$6$aReallyHardSalt$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJm19be.cfi3/qxIf.
hsGpS41BqMhSrHVXgMpdjS6xeKZAs02.
e5d8870e5bdd26602cab8dbe07a942c8669e56d6
```

3.2 Step 1: CrackStation

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

```
F09EDCB1FCEFC6DFB23DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85
1DFECA0C002AE40B8619ECF94819CC1B
$6$aReallyHardSalt$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJm19be.cfi3/qxIf.hs
GpS41BqMhSrHVXgMpdjS6xeKZAs02.
e5d8870e5bdd26602cab8dbe07a942c8669e56d6
```



[Crack Hashes](#)

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, rpeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
F09EDCB1FCEFC6DFB23DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85	sha256	Found
1DFECA0C002AE40B8619ECF94819CC1B	NTLM	Found
\$6\$aReallyHardSalt\$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJm19be.cfi3/qxIf.hsGpS41BqMhSrHVXgMpdjS6xeKZAs02.	Unknown	Unrecognized hash format.
e5d8870e5bdd26602cab8dbe07a942c8669e56d6	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

Figure 3: CrackStation results for Task 2

- First hash: SHA-256
- Second hash: NTLM
- Third hash: SHA512crypt
- Fourth hash: SHA-1

3.3 Step 2: Prepare Files

```
echo '$6$aReallyHardSalt$6WKUTqzq.UQQmrm0p/T7MPpMbGNnzXPMAXi4bJm19be.cfi3/qxIf.
hsGpS41BqMhSrHVXgMpdjS6xeKZAs02.' > sha512.txt

echo 'e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme' > sha1.txt
```

3.4 Step 3: Crack with Hashcat

SHA512crypt:

```
hashcat -m 1800 -a 0 sha512.txt /usr/share/wordlists/rockyou.txt
```



```
$ hashcat -m 1800 -a 0 sha512.txt /usr/share/wordlists/rockyou.txt --show
$6$aReallyHardSalt$6WKUTqzq.UQqrm0p/T7MPpMbGNzXPMAXi4bJm19be.cfi3/qxIf.hsGpS418qMhSrHVXgMpdjS6xeKZAs02.:w
```

Figure 4: Hashcat cracking SHA512crypt

SHA1:

```
hashcat -m 160 -a 0 sha1.txt /usr/share/wordlists/rockyou.txt -O -w 4
```



```
$ hashcat -m 160 -a 0 sha1.txt /usr/share/wordlists/rockyou.txt -O -w 4 --show
e5d8870e5bdd26602cab8dbe07a942c8669e56d6:tryhackme:4
```

Figure 5: Hashcat cracking SHA1

4 Conclusion

This is my writeup for THM-CrackTheHash, thanks for reading. – Zackary Jordan